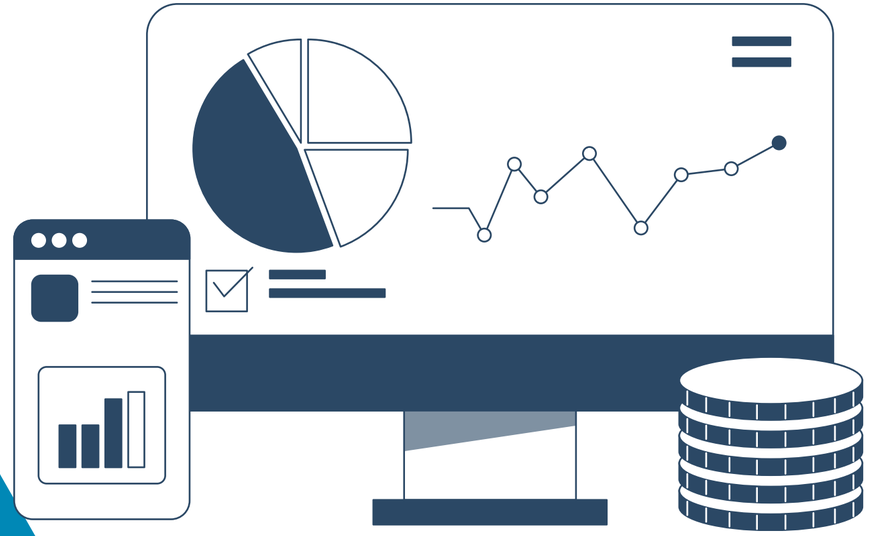
DATA ANALYTICS

30 days



DAY 24

Agenda

- Select specific columns from a dataset
- Filter records based on conditions
- Apply multiple filtering conditions
- Sort data for ranking and analysis
- Create new calculated columns
- Perform aggregation operations
- Use Group By for business analysis
- Extract meaningful business insights

Problem Statement

Management wants to know:

- Which city generates the highest revenue?
- Who are the top customers?
- Which region performs the best?
- Which customers generate the highest profit?

Selecting Columns

Why Select Columns?

- Focus only on relevant information
- Improve readability
- Reduce unnecessary data processing

```
df[['Customer', 'Revenue']]
```

Filtering Data

Management wants to see only high-value sales.

```
df[df['Revenue'] > 5000]
```

```
df[(df['City'] == 'Dhaka') &
   (df['Revenue'] > 5000)] # multiple condition
```

Sorting Data

Rank customers by revenue.

```
df.sort_values(
  'Revenue',
  ascending=False
)
```

Creating New Columns

Revenue and Profit are available, but Profit Percentage is not.

```
Profit Percentage =
(Profit / Revenue) × 100
```

```
df['Profit_Percentage'] = (
    df['Profit'] /
    df['Revenue']
) * 100
```

Aggregation Functions

Summarize data using statistical calculations.

```
df['Revenue'].sum()

df['Revenue'].mean()

df['Revenue'].max()

df['Revenue'].min()
```

Understanding Group By

```
df.groupby('City')['Revenue'].sum()
```

Business Question

Which City Generates the Highest Revenue?

```
df.groupby('City')['Revenue'] \
  .sum() \
  .sort_values(
      ascending=False
  )
```

Who Are the Top Revenue Customers?

```
df.sort_values(
    'Revenue',
    ascending=False
).head(5)
```

Business Question

Who Is the Most Profitable Customer?

```
df.sort_values(
    'Profit',
    ascending=False
).head(1)
```

Which Region Performs Best?

```
df.groupby(
    'Region'
)['Revenue'].sum()
```

Complete Example

Data Manipulation Skills

- Selecting Columns
- Filtering Data
- Applying Multiple Conditions
- Sorting Records
- Creating New Columns

Data Transformation Skills

- Aggregation
- Group By Analysis
- Business Insight Generation

Practice Assignment

- **Find all customers with Revenue greater than 10,000.**
- **Create a new column called Profit_Percentage.**
- **Calculate the total Revenue for each City.**
- **Identify the Top 3 customers based on Revenue.**
- **Determine the highest revenue-generating Region.**

THANK YOU

FOR YOUR ATTENTION